

18-07-2026

Agenda: ML-20

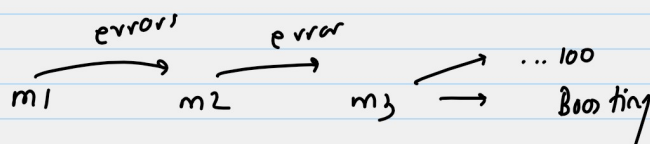
Gradient Boosting } Regression •

Xgboost

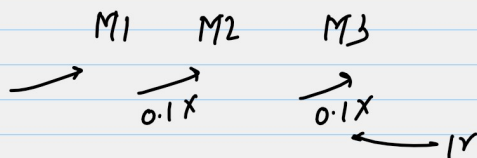
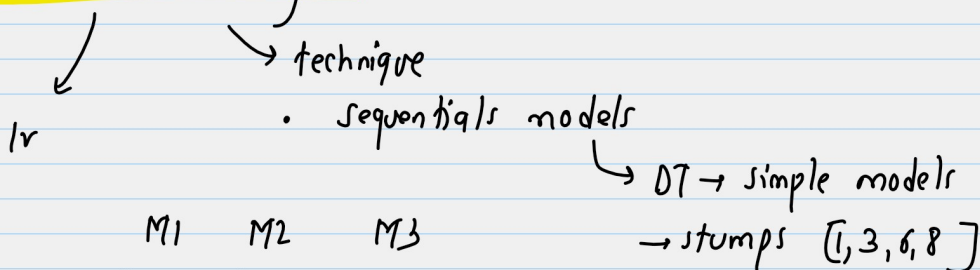
practicals + docker intro

Team will upload
wednesday still recording
soon. Sorry for the day

please read this



Gradient Boosting



$$\text{loss function} = \frac{1}{2} (y - \hat{y})^2$$

Dataset

ID	Size	Beds	price (y)
1	800	2	120
2	900	2	130
3	1000	3	150
4	1100	3	170
5	1200	4	200

Step: 0 $\rightarrow$ Initial model (f_0)

$$f0 = \text{mean}(y) \rightarrow (120 + 130 + 150 + 170 + 200) / 5 = 154$$

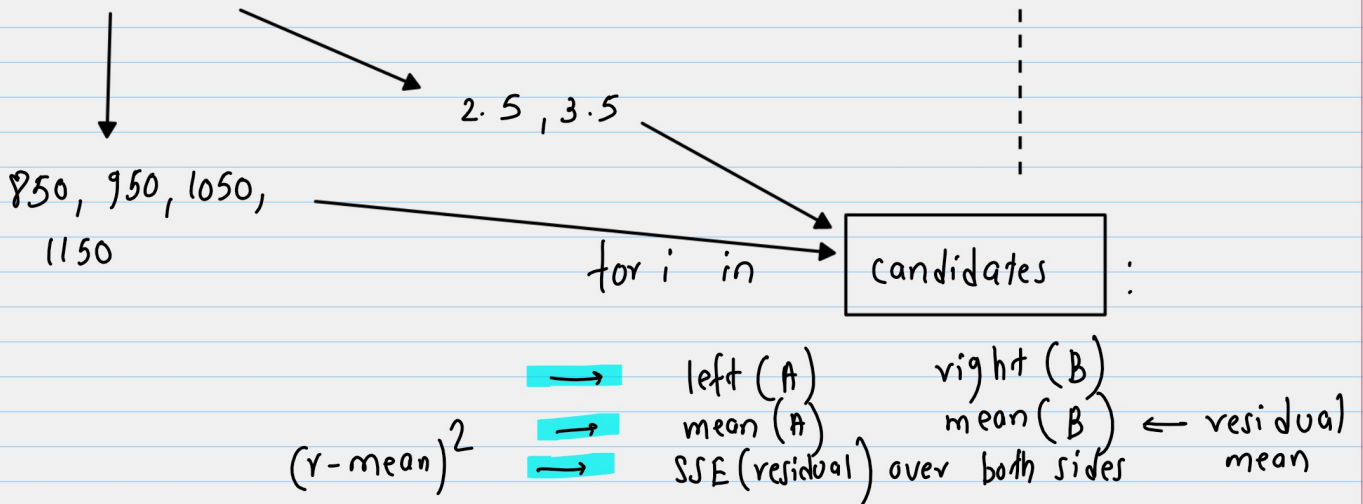
→ calculate residuals $(y - f) = (y - f_0)$

ID	Size	Beds	price (y)	$r_1(y - f_0)$
1	800	2	120	-34
2	900	2	130	-24
3	1000	3	150	-4
4	1100	3	170	+16
5	1200	4	200	+46

target for the stump

→ candidate splits

ID	Size	Beds	price (y)	$r_1(y - f_0)$
1	800	2	120	-34
2	900	2	130	-24
3	1000	3	150	-4
4	1100	3	170	+16
5	1200	4	200	+46



Feature	Threshold	left (A)	Right (B)	mean (A)	mean (B)	SSE
Size	1050	[1, 2, 3]	[4, 5]	-20.66	31	916 ← lowest
Size	950	[1, 2]	[3, 4, 5]	-29	19.33	1316.66
Size	1150	[1, 2, 3, 4]	[5]	-11.5	46	1475
Size	850	[1]	[2, 3, 4, 5]	-34	8.5	2675
Beds	2.5	[1, 2]	[3, 4, 5]	-29	19.33	1316.66
Beds	3.5	[1, 2, 3, 4]	[5]	-11.5	46	1475

Size < 1050 chosen as stump

left → right

Full calculation example below

ID	Size	residual	ID	Size	residual
1	800	-34	4	1100	16
2	900	-24	5	1200	46
3	1000	-4			

mean = -20.66

mean = 31

← from above table

$SSE(A) =$

$SSE(B)$

$$\begin{aligned}
 (-34 - (-20.66)) &= -13.33 \rightarrow \text{squared} = 177 \\
 (-24 - (-20.66)) &= -3.33 \rightarrow = 11.1 \\
 (-4 - (-20.66)) &= 16.66 \rightarrow = 277.77 \\
 \hline
 &466
 \end{aligned}$$

$$\begin{aligned}
 (16, 31) &= -15 \rightarrow \text{squared} = 225 \\
 (46, 31) &= 15 \rightarrow 225 \\
 \hline
 &450
 \end{aligned}$$

$$\begin{aligned}
 \text{Total SSE} &= SSE(A) + SSE(B) \\
 &= 916.66
 \end{aligned}$$

$f_0(x) \rightarrow 154$

← model-1

$h_1(x) \rightarrow$
 Size ≤ 1050
 ↙ ↘
 -20.66 31

← model-2

$$F_1(x) = f_0 + r * h_1(x)$$

ID	Size	Beds	price (y)	$r_1(y - f_0)$	f_1	$r_2(y - f_1)$
1	800	2	120	-34	151.93	-31.93
2	900	2	130	-24	151.93	-21.93
3	1000	3	150	-4	151.93	-1.93
4	1100	3	170	+16	157.1	12.90
5	1200	4	200	+46	157.1	42.90

compute f_1 :

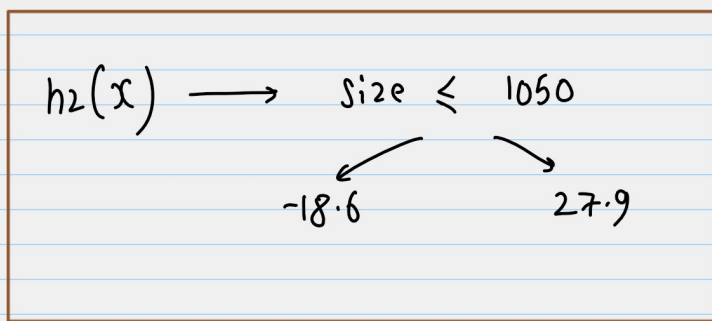
$$10-1(A) : 154 + 0.1 \times (-20.66) = 151.93$$

$\uparrow$
 $800 \leq 1050$
 $\swarrow \quad \searrow$
 $-20.66 \quad 31.0$

we will use r_2 as the next stump

Feature	Threshold	left(A)	right(B)	mean(A)	mean(B)	SSE
Size	1050	[1,2,3]	[4,5]			
Size	950	[1,2]	[3,4,5]			
Size	1150	[1,2,3,4]	[5]			
Size	850	[1]	[2,3,4,5]			
Beds	2.5	[1,2]	[3,4,5]			
Beds	3.5	[1,2,3,4]	[5]			

Feature	Threshold	left(A)	right(B)	mean(A)	mean(B)	SSE
Size	1050	[1,2,3]	[4,5]	-18.60	27.90	916.66 ← lowest
Size	950	[1,2]	[3,4,5]	-26.93	17.95	1093.35
Size	1150	[1,2,3,4]	[5]	-10.72	42.9	1210.85
Size	850	[1]	[2,3,4,5]	-31.93	8.5	2236.69
Beds	2.5	[1,2]	[3,4,5]	-26.93	17.95	1093.35
Beds	3.5	[1,2,3,4]	[5]	-10.72	42.9	1210.85



→ new stump

→ model-3

$$f_1(x) = f_0 + r * h_1(x)$$

$$f_2(x) = f_0 + r * h_1(x) + r * h_2(x)$$

$$= f_1(x) + r * h_2(x)$$

↙ copied from above for f_1 value

ID	Size	Beds	price (y)	$r_1(y - f_0)$	f_1	$r_2(y - f_1)$
1	800	2	120	-34	151.93	-31.93
2	900	2	130	-24	151.93	-21.93
3	1000	3	150	-4	151.93	-1.93
4	1100	3	170	+16	157.1	+2.90
5	1200	4	200	+46	157.1	+2.90

ID	Size	Beds	price (y)	$r_1(y - f_0)$	f_1	$r_2(y - f_1)$	f_2	$r_3(y - f_2)$
1	800	2	120	-34	151.93	-31.93	150.07	-30.07
2	900	2	130	-24	151.93	-21.93	150.07	-20.07
3	1000	3	150	-4	151.93	-1.93	150.07	-0.07
4	1100	3	170	+16	157.1	+2.90	159.89	+10.11
5	1200	4	200	+46	157.1	+2.90	159.89	+40.11

compute $-f_2$

$$ID = f_1(x) + r * h_2(x)$$

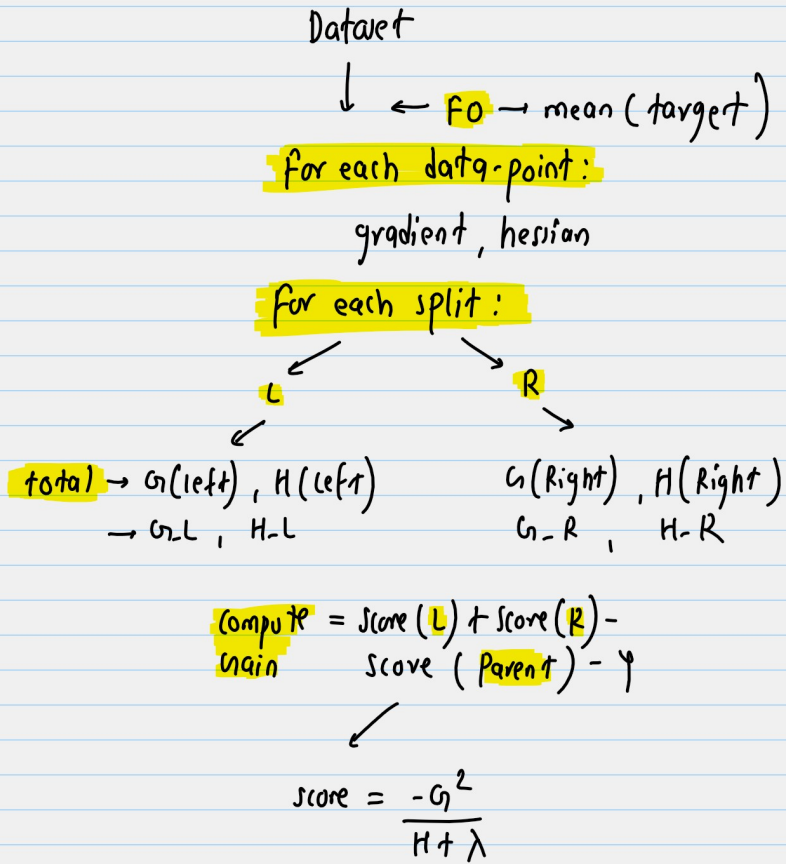
$$[1,2,3] = 151.93 + 0.1 * (-18.6) = 150.07$$

$$ID = 157.1 + 0.1 * 27.9 = 159.89$$

Because the residuals maintain the same sign pattern (IDs 1-3 negative, 4-5 positive), splitting along Size still best separates negative vs positive residuals and thus minimizes SSE. As iterations progress residual magnitudes shrink, and at some point another split (e.g., Beds or a different Size cutoff) might become best, but early on Size is the dominant predictive feature.

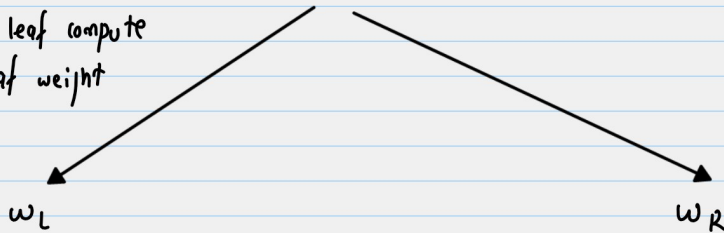
Xg boost

Xtreme - gradient boost



choose-split with highest possible gain

for each leaf compute final leaf weight



Final prediction $(f_{t+1}) = f_0 + \eta \cdot f_t(x)$

$g_B \leftarrow x_{gB}$

loss = squared error

$$l = \frac{1}{2} (y - \hat{y})^2$$

$$\text{gradient} = \frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

$$\text{hessian} = \frac{\partial^2 L}{\partial y^2} \approx 1$$

$$\lambda(\text{regularization}) = 1$$

$$\Psi(\text{minimum gain to split}) = 0$$

$\lambda \rightarrow$ denominator (L2)
 $\rightarrow$ shrinks core

$\gamma \rightarrow$ subtraction penalty (1)

$\alpha \rightarrow$ leaf weight calc,
not split cost (1r)

$$\omega = \frac{-G}{H + \lambda}$$

$\eta = 1$
for simple calculation

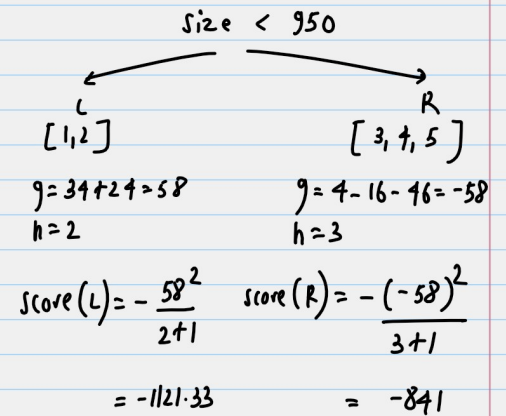
ID	Size	Beds	price (y)	$f_0(\hat{y}_i^{(0)})$	$g = \hat{y}_0 - y$	h
1	800	2	120	154	34	1
2	900	2	130	154	24	1
3	1000	3	150	154	4	1
4	1100	3	170	154	-16	1
5	1200	4	200	154	-46	1

Iteration-1 → Build Tree

Size	850	gain	1156
Size	950	1962.33	
Size	1050	707	
Beds	2.5	1156	
Beds	3.5	707	

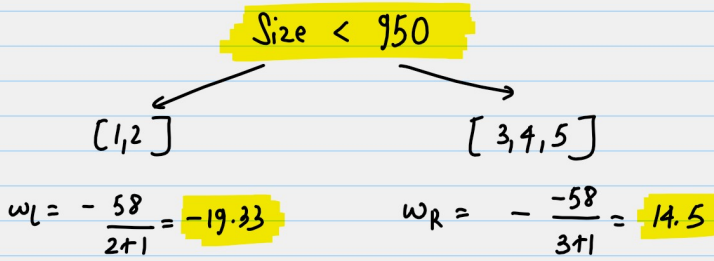
highest gain

calculation



leaf-weights:

$$w = -\frac{g}{H + \lambda}$$



score(Parent)

$$g = 34 + 24 + 4 - 16 - 46 = 0$$

$$h = 5$$

$$= -\frac{0^2}{5+1} = 0$$

$$\text{Gain} = 1121.33 + 841 = 1962.33$$

$$(f_{t+1}) = f_0 + \eta \cdot f_t(x)$$

ID	Size	Beds	price (y)	$f_0(\hat{y}_i^{(0)})$	leaf	weight	F_i
1	800	2	120	154	L	-19.33	134.67
2	900	2	130	154	L	-19.33	134.67
3	1000	3	150	154	R	+14.5	168.5
4	1100	3	170	154	R	+14.5	168.5
5	1200	4	200	154	R	+14.5	168.5

$$10-1 \rightarrow 154 + 1 * (-19.33) = 134$$

Iteration-2

ID	Size	Beds	price (y)	F_i	g	h
1	800	2	120	134.67	14.67	1
2	900	2	130	134.67	4.67	1
3	1000	3	150	168.5	18.5	1
4	1100	3	170	168.5	-1.5	1
5	1200	4	200	168.5	-31.5	1

Gradient-Boost

Xgboost

Use derivative of loss

Yes

Yes

Use second order derivative

No

Yes

split score

SSE

Based on gain
(g, h, λ)

Regularization

weak (α)

Strong (λ, α, γ)

Tree complexity control

max-depth

prunes automatically
↑
max-depth

leaf value

mean residual

$$-\frac{g}{H + \lambda}$$

system speed

slower

much faster, parallel